

[1] Index

Rev	Date	Designer	Description
A	25.01.26	Ganghyeok Lim	Create design project

Index

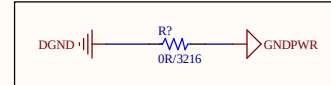
#1 Index	#8 ??
#2 Overview	#9 ??
#3 ??	
#4 ??	
#5 ??	
#6 ??	
#7 ??	

Title Sitara_Project_Rev_A.PrjPcb		Rev A
Doc Index.SchDoc		 
Sheet # 1 of 12	Engineer Ganghyeok Lim	
Date 2025-01-26		
This document is copyright of ArkX and shall not be revealed, produced, copied, in whole or in part, nor used for any purpose other than submitted.		

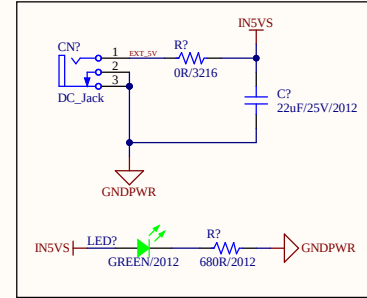
	1	2	3	4																												
A	Sitara Project B'd (Rev.A) [2] Overview																															
B																																
C																																
D	<table><tr><td colspan="3">Title</td><td>Rev</td></tr><tr><td colspan="3">Sitara_Project_Rev_A.PrjPcb</td><td>A</td></tr><tr><td colspan="2">Doc</td><td colspan="2">Overview.SchDoc</td></tr><tr><td>Sheet # 2 of 12</td><td colspan="3">Engineer</td></tr><tr><td></td><td colspan="3">Ganghyeok Lim</td></tr><tr><td>Date</td><td colspan="3">2025-01-26</td></tr><tr><td colspan="4">This document is copyright of ArkX and shall not be revealed, produced, copied, in whole or in part, nor used for any purpose other than submitted.</td></tr></table>				Title			Rev	Sitara_Project_Rev_A.PrjPcb			A	Doc		Overview.SchDoc		Sheet # 2 of 12	Engineer				Ganghyeok Lim			Date	2025-01-26			This document is copyright of ArkX and shall not be revealed, produced, copied, in whole or in part, nor used for any purpose other than submitted.			
Title			Rev																													
Sitara_Project_Rev_A.PrjPcb			A																													
Doc		Overview.SchDoc																														
Sheet # 2 of 12	Engineer																															
	Ganghyeok Lim																															
Date	2025-01-26																															
This document is copyright of ArkX and shall not be revealed, produced, copied, in whole or in part, nor used for any purpose other than submitted.																																
	1	2	3	4																												

[3] Power

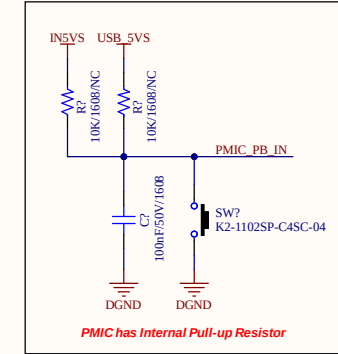
Digital GND



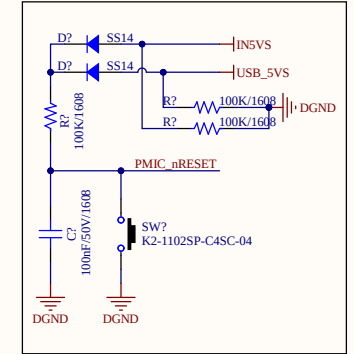
Power Connector (5V)



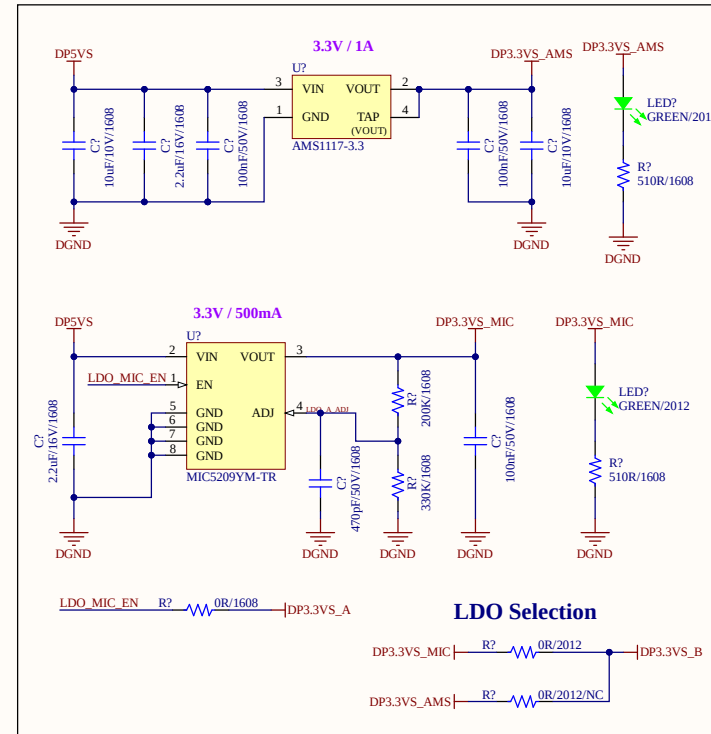
PMIC Power-On Button



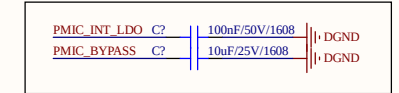
PMIC Reset Button



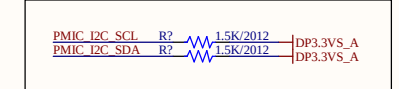
LDO (3.3V / 500mA)



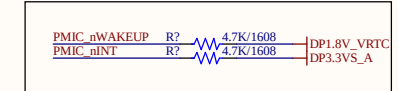
INT LDO / BYPASS Pin



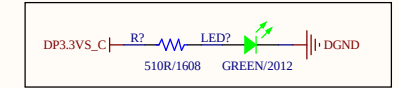
I2C Pull-up Resistor



nWAKE-UP / nINT Pull-up



PMIC Power LED



LDO Selection

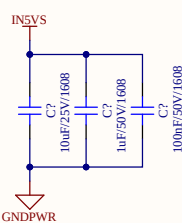


PMIC's Output Voltage based on AM3358's OPP100

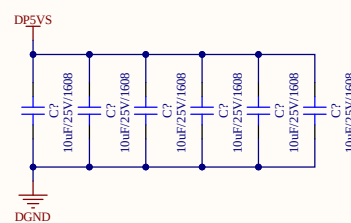
- DCDC1 (1.5V / 1.2A) => CPU's "VDDS_DDR"
- DCDC2 (1.1V / 1.2A) => CPU's "VDDS_MPU"
- DCDC3 (1.1V / 1.2A) => CPU's "VDDS_CORE"
- LDO1 (1.8V / 0.1A) => CPU's "VIO" & "VDDS" & "VDDS_RTC"
- LDO2 (3.3V / 0.1A) => No Connection with CPU
- LDO3 (1.8V / 0.4A) => Some Connection with CPU
- LDO4 (3.3V / 0.4A) => Some Connection with CPU

Decoupling Capacitors

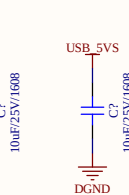
Decap for IN5VS



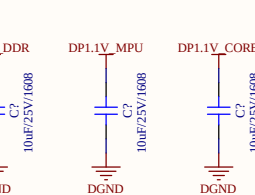
Decap for DP5VS



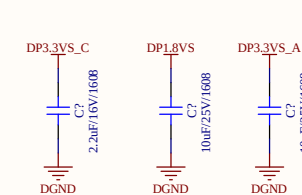
Decap for USB_5VS



Decap for DCDC 1/2/3

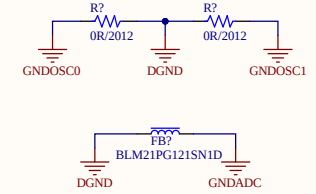
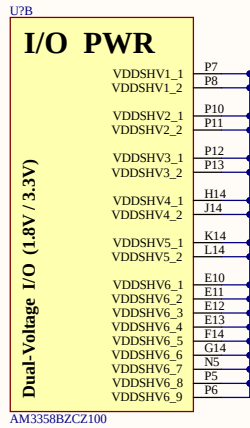


Decap for LDO 1~4

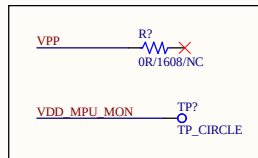
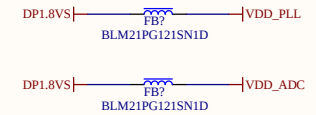


Title	Sitara_Project_Rev_A.PrjPcb		Rev	*
Doc	Power.SchDoc			
Sheet # 3 of 12	Engineer *			
Date *				
This document is copyright of ArkX and shall not be revealed, produced, copied, in whole or in part, nor used for any purpose other than submitted.				

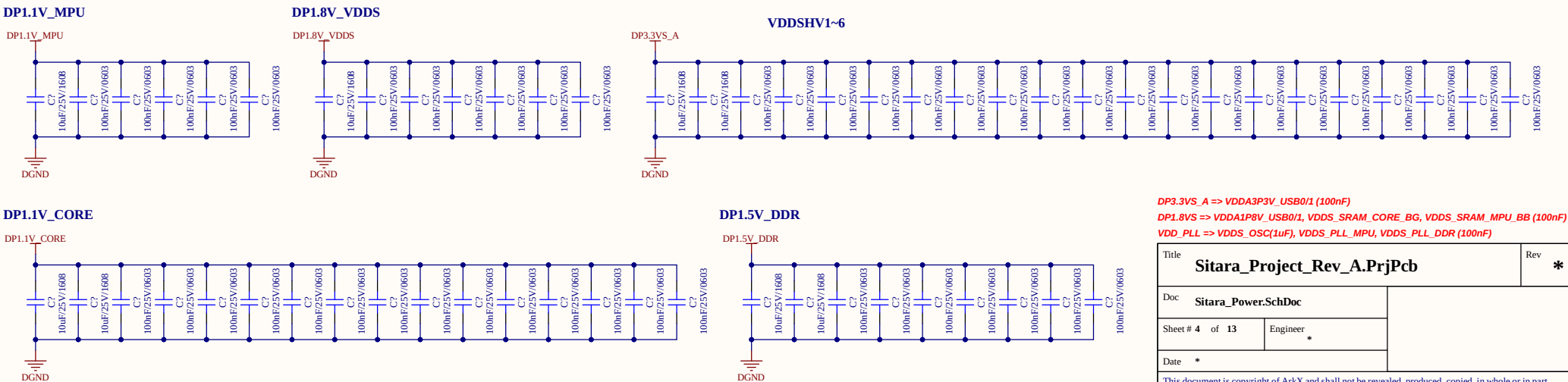
Place Decaps Closely to Each Pin of PMIC



Ferrite Bead Connection



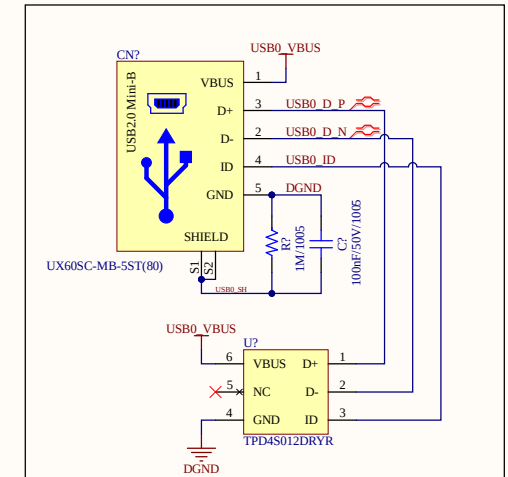
Decoupling Capacitors



Title		Sitaro_Project_Rev_A.PrjPcb		Rev	*
Doc		Sitaro_Power.SchDoc			
Sheet # 4 of 13		Engineer *			
Date *					
This document is copyright of ArkX and shall not be revealed, produced, copied, in whole or in part, nor used for any purpose other than submitted.					

[illegible]

USB_5VS — R? — USB0_VBUS
0R/3216



BC For LED

USB0_VBUS

LED?

GREEN/2012

R?

1K/1608

DGND

USB0_VBUS

C?

4.7uF/10V/1005

DGND

RTC_KALDO_nEN R? 10K/1608 DGND

DP3.3VS_A

470R/1608

LED?

BLUE/2012

Q?

2N7002

DGND

AM3 USER_LED1

470R/1608

LED?

BLUE/2012

Q?

2N7002

DGND

AM3 USER_LED2

470R/1608

LED?

BLUE/2012

Q?

2N7002

DGND

AM3 USER_LED3

470R/1608

LED?

BLUE/2012

Q?

2N7002

DGND

AM3 USER_LED4

470R/1608

LED?

BLUE/2012

Q?

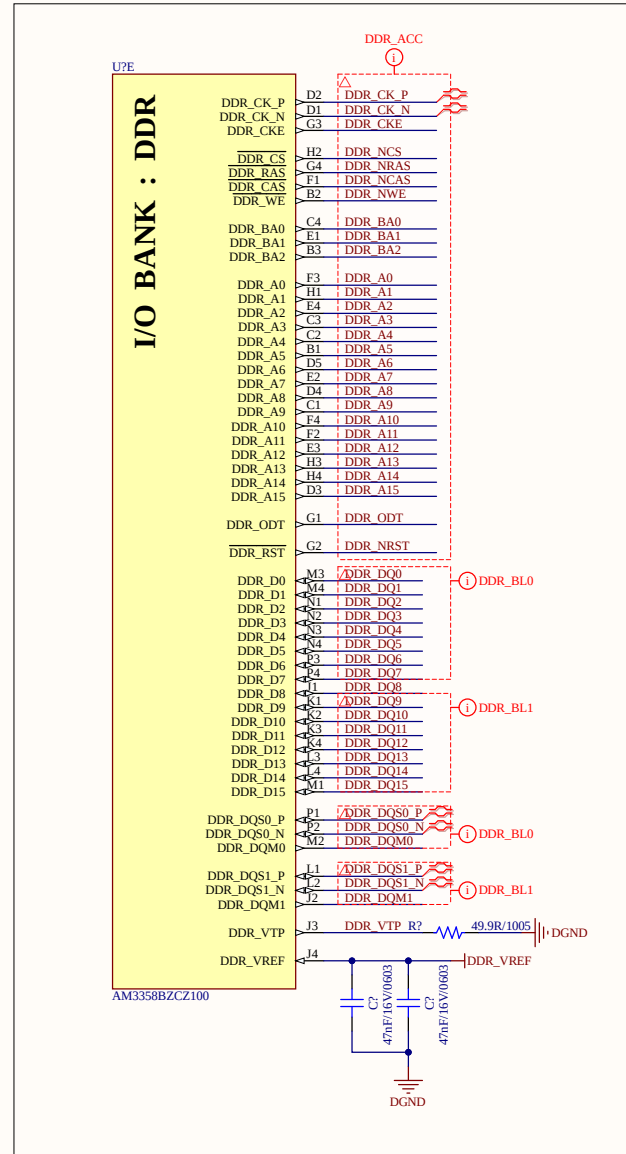
2N7002

DGND

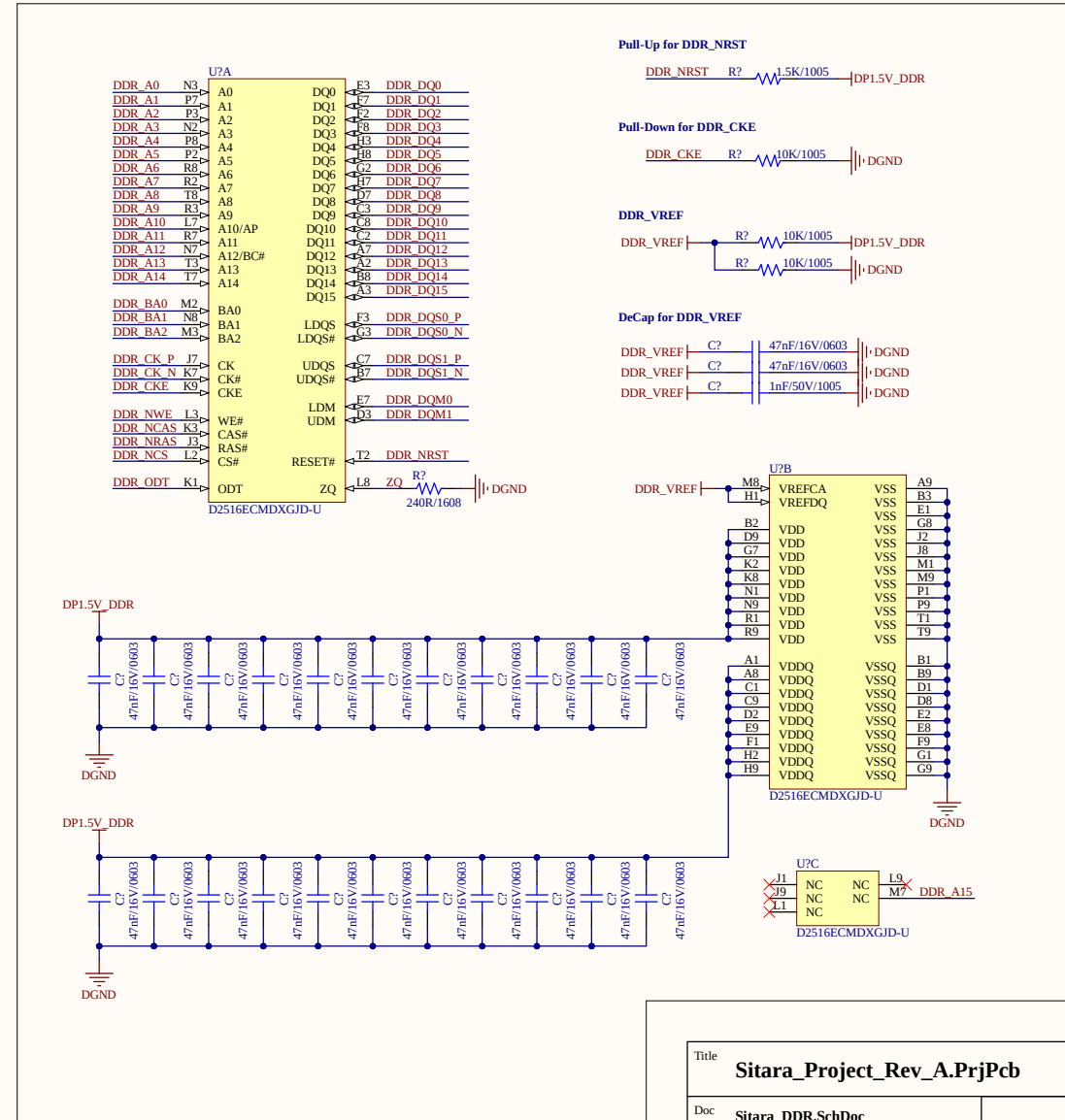
Title		Sitara_Project_Rev_A.PrjPcb		Rev	*
Doc		Sitara_AUX.SchDoc			
Sheet # 5	of 12	Engineer	*		
Date		*			
This document is copyright of ArkX and shall not be revealed, produced, copied, in whole or in part, nor used for any purpose other than submitted.					

[6] Sitara DDR Interface

Sitara DDR Memory Controller Interface (EMIF)



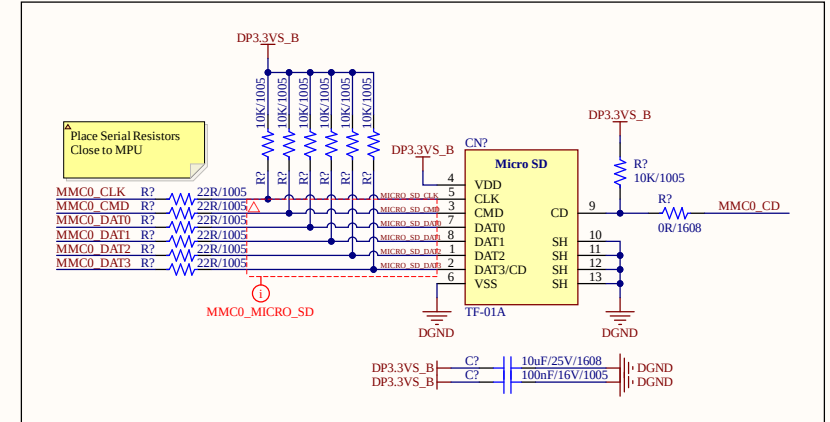
DDR3L Memory



Title		Rev
Sitara_Project_Rev_A.PrjPcb		*
Doc		Sitara_DDR.SchDoc
Sheet # 6 of 12	Engineer *	
Date *		
This document is copyright of ArkX and shall not be revealed, produced, copied, in whole or in part, nor used for any purpose other than submitted.		

MICRO SD Card (MMC0)

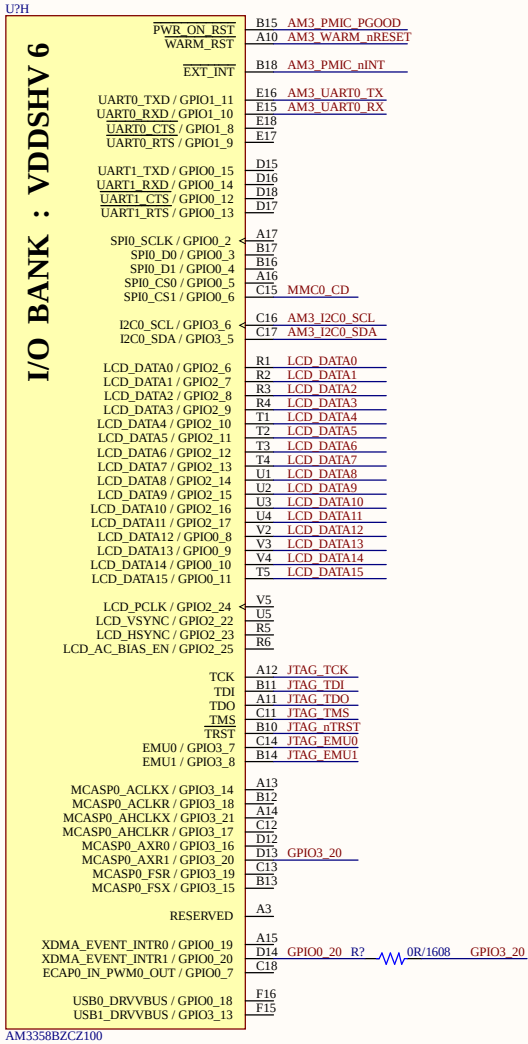
The diagram illustrates the connection of an AT24C16 EEPROM to an I2C bus. The chip is represented by a yellow box with pins 1 through 8. Pin 1 is VCC, pin 2 is SCL, pin 3 is SDA, pin 4 is GND, pin 5 is WP, pin 6 is NC, pin 7 is NC, and pin 8 is NC. The chip is connected to a DP3.3V5_A supply. The I2C address is 1010_XXXX. The circuit includes pull-up resistors (R1, R2, R3) connected to the SCL, SDA, and WP lines. The WP line is also connected to a 100nF/25V/0603 capacitor and a DGND ground.



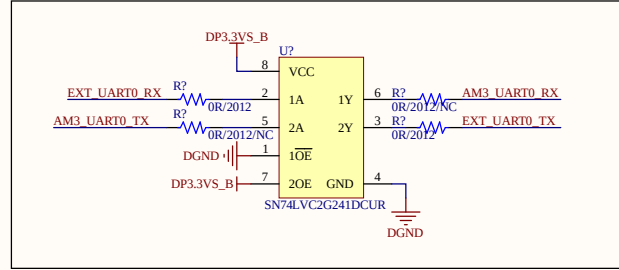
K7	RF14	NC61	J1
K6	RF15	RF14	H4
K5	RF12	NC59	H3
K4	RF11	NC58	H2
K3	VSF7	NC57	H1
K2	VSF6	NC55	HK
K1	VSF5	NC54	G4
J10	VSF3	NC53	G3
J9	VSF2	NC51	G2
J8	VSF1	NC50	G1
J7	NC109	NC49	U1
J6	NC108	NC48	U2
J5	NC107	NC47	F2
J4	NC106	NC46	F1
J3	NC105	NC44	F2
J2	NC104	NC43	F1
J1	NC103	NC42	U1
I10	NC102	NC41	U2
I9	NC101	NC40	F2
I8	NC100	NC39	F1
I7	NC99	NC38	U1
I6	NC97	NC37	U2
I5	NC96	NC35	D4
I4	NC95	NC34	D3
I3	NC94	NC33	D2
I2	NC93	NC32	D1
I1	NC90	NC31	D1
H10	NC89	NC29	C4
H9	NC88	NC28	C3
H8	NC87	NC26	C2
H7	NC86	NC25	C1
H6	NC85	NC24	C1
H5	NC84	NC23	C1
H4	NC83	NC22	C1
H3	NC82	NC21	C5
H2	NC81	NC20	C3
H1	NC80	NC19	U1
G10	NC79	NC18	U2
G9	NC78	NC17	B2
G8	NC77	NC16	B1
G7	NC76	NC15	B1
G6	NC75	NC14	B0
G5	NC74	NC13	B4
G4	NC73	NC12	B3
G3	NC72	NC11	B1
G2	NC71	NC10	A4
G1	NC70	NC9	A3
F10	NC69	NC8	A2
F9	NC68	NC7	A1
F8	NC67	NC6	A0
F7	NC66	NC5	A0
F6	NC65	NC4	A3
F5	NC64	NC3	A3
F4	NC63	NC2	A1
F3	NC62	NC1	A1

Title		Sitara_Project_Rev_A.PrjPcb		Rev	*
Doc		Sitara_IO_Bank_123.SchDoc			
Sheet # 7 of 12		Engineer *			
Date *					
This document is copyright of ArkX and shall not be revealed, produced, copied, in whole or in part, nor used for any purpose other than submitted.					

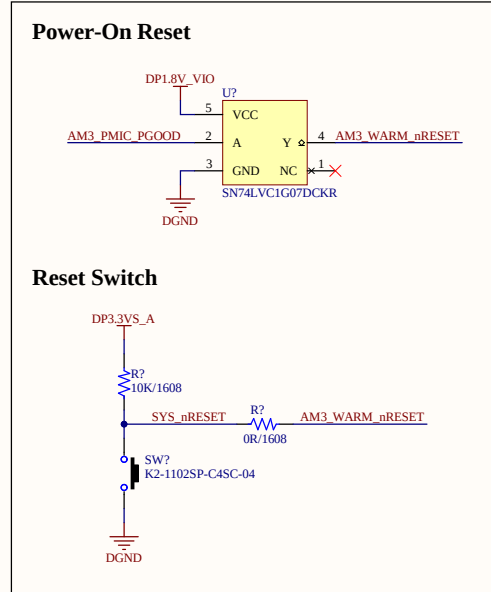
[9] Sitara I/O Bank 6



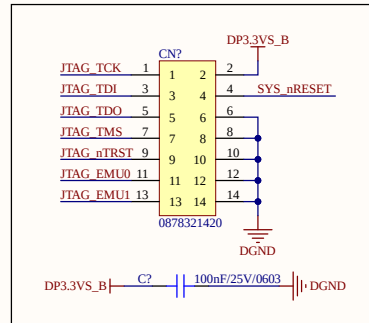
UART0 External Connector



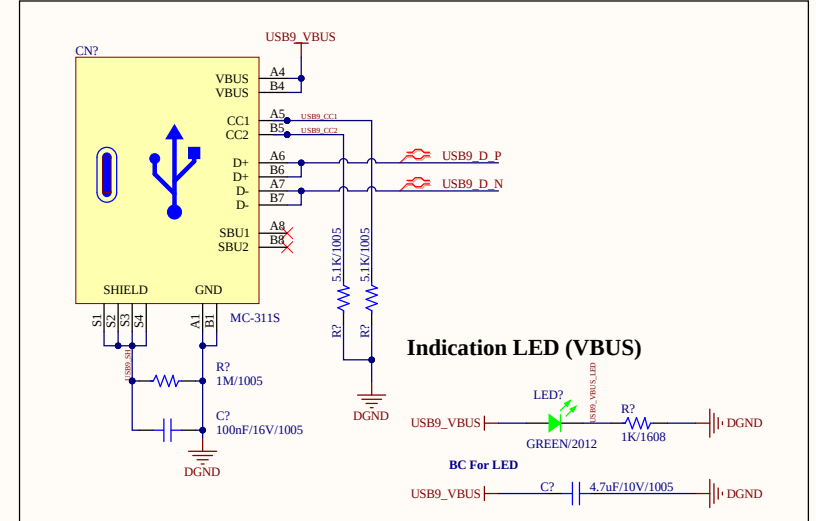
System Reset



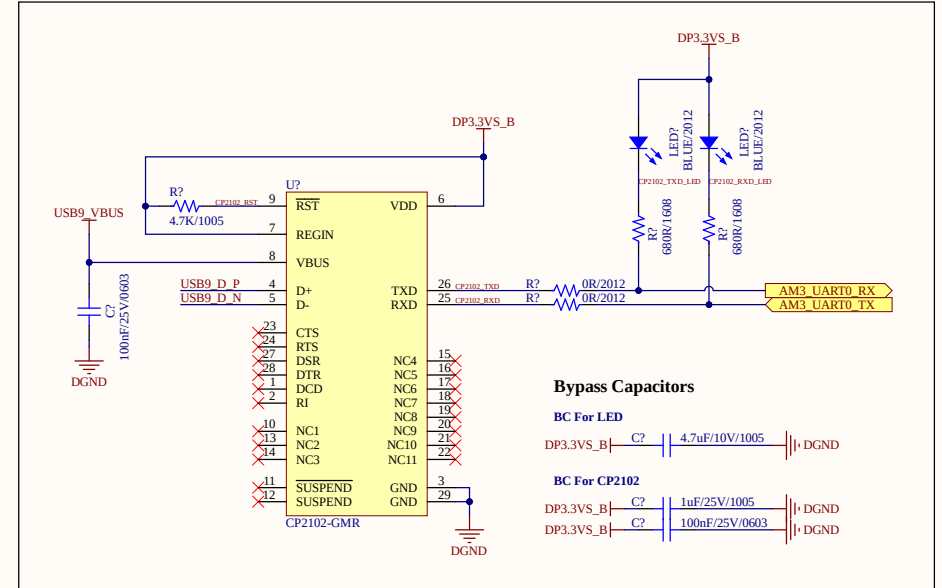
JTAG Connector



USB Connector (Type-C)



USB - UART Bridge



Title		Rev
Sitara_Project_Rev_A.PrjPcb		*
Doc		Sitara_IO_Bank_6.SchDoc
Sheet # 9 of 12	Engineer *	
Date *		
This document is copyright of ArkX and shall not be revealed, produced, copied, in whole or in part, nor used for any purpose other than submitted.		

	1	2	3	4																										
A	Sitara Project B'd (Rev.A) [?] eMMC Module																													
B																														
C																														
D	<table><tr><td colspan="3">Title</td><td colspan="1">Rev</td></tr><tr><td colspan="3">Sitara_Project_Rev_A.PrjPcb</td><td>*</td></tr><tr><td colspan="2">Doc</td><td colspan="2">eMMC_Module.SchDoc</td></tr><tr><td colspan="1">Sheet # 10 of 13</td><td colspan="1">Engineer</td><td colspan="2" rowspan="2"></td></tr><tr><td colspan="1"></td><td colspan="1">*</td></tr><tr><td colspan="2">Date</td><td colspan="2">*</td></tr><tr><td colspan="4">This document is copyright of ArkX and shall not be revealed, produced, copied, in whole or in part, nor used for any purpose other than submitted.</td></tr></table>				Title			Rev	Sitara_Project_Rev_A.PrjPcb			*	Doc		eMMC_Module.SchDoc		Sheet # 10 of 13	Engineer				*	Date		*		This document is copyright of ArkX and shall not be revealed, produced, copied, in whole or in part, nor used for any purpose other than submitted.			
Title			Rev																											
Sitara_Project_Rev_A.PrjPcb			*																											
Doc		eMMC_Module.SchDoc																												
Sheet # 10 of 13	Engineer																													
	*																													
Date		*																												
This document is copyright of ArkX and shall not be revealed, produced, copied, in whole or in part, nor used for any purpose other than submitted.																														
	1	2	3	4																										

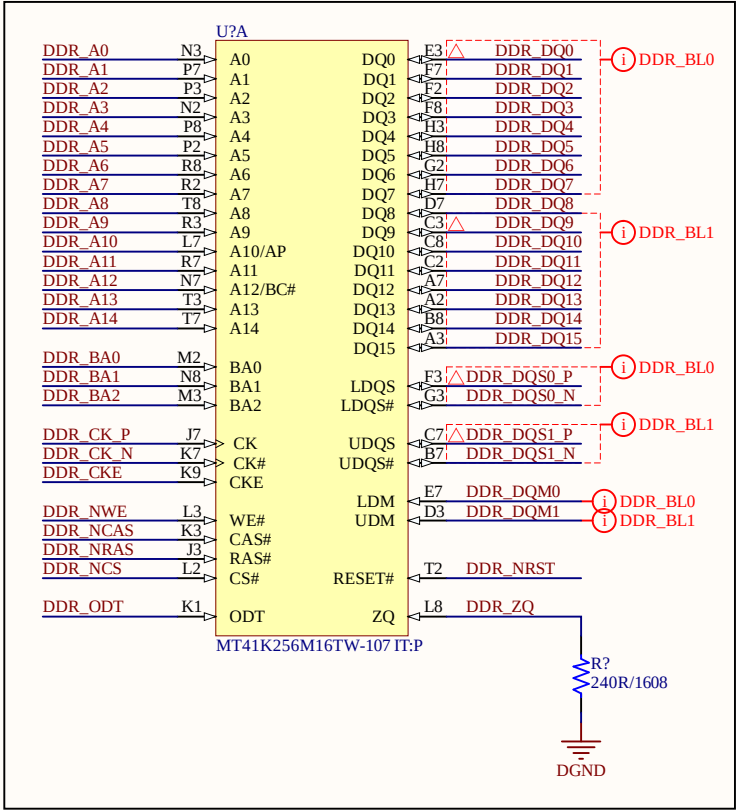
[?] Schematic Design Check-List

#1 Index

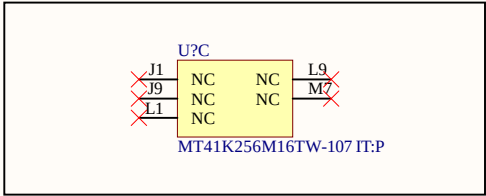
No.	Category	Item	Description	Confirm

[7] DDR3L Memory Module

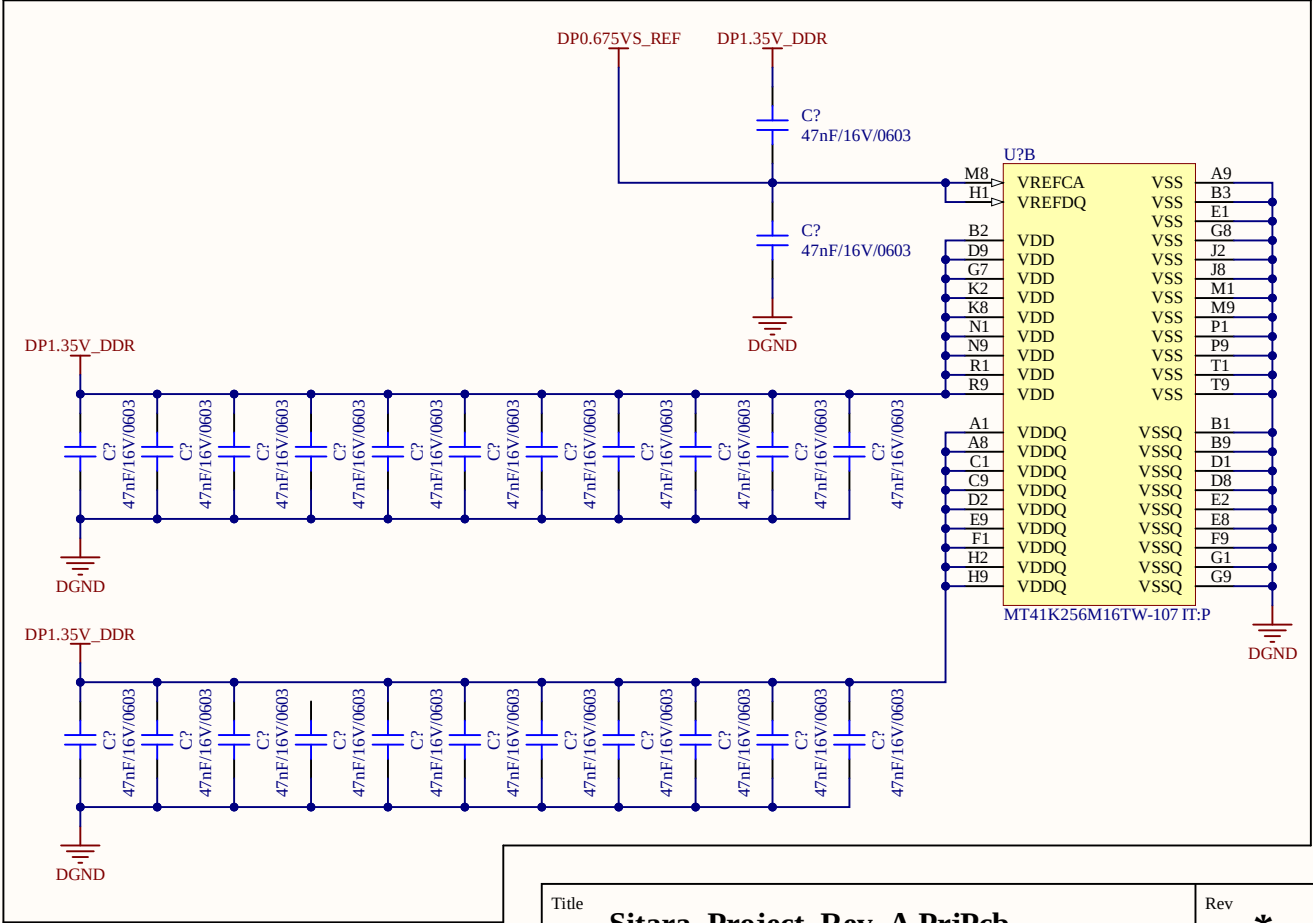
ACC & DATA Connections



Unused Pin



DDR Power & Decoupling



Title Sitara_Project_Rev_A.PrjPcb		Rev *
Doc DDR3L_Module.SchDoc		
Sheet # 12 of 13	Engineer *	
Date *		
This document is copyright of ArkX and shall not be revealed, produced, copied, in whole or in part, nor used for any purpose other than submitted.		

